

continuous probability distributions RANDOM VARIABLES WITH CONTINUOUS DOMAIN

- **density** function $f(x)$:

$$1) f(x) \geq 0, f(x) \leq 1,$$

$$\Omega = (-\infty, +\infty)$$

$$\Omega = (0, +\infty)$$

...

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

TAKES AS INPUT A DOMAIN VALUE x
RETURNS AS OUTPUT THE DEGREE OF
CONFIDENCE OF x : $f(x)$

- **cumulative** function $F(x)$:

$$F(x) = \int_{-\infty}^x f(y) dy$$

$F(x)$ PROBABILITY OF
OBSERVING AT MOST x

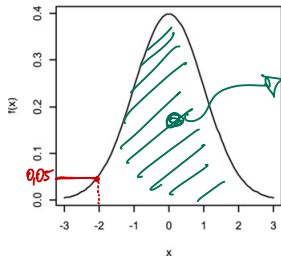
what is the probability that...

... $x = a$? $f(a)$

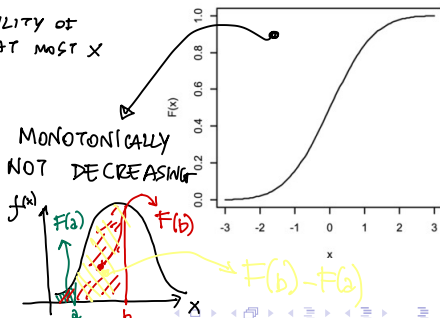
... $x \leq b$? $F(b)$

... $a \leq x \leq b$? $F(b) - F(a)$

Probability density function



Cumulative distribution function



given a continuous random variable X with values x defined on $[-\infty, \infty]$ associated to a continuous probability density function $f(x)$

- MEAN, EXPECTED VALUE, PREDICTION

$$m = \mathbb{P}(X) = \int_{-\infty}^{\infty} xf(x)dx$$

↓
WEIGHTED
SUM → $f(x)$ IS THE
WEIGHT

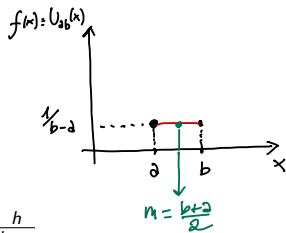
- VARIANCE

$$\text{var}(X) = \int_{-\infty}^{\infty} (x - m)^2 f(x) dx$$

UNIFORM DISTRIBUTION $U_{a,b}(x)$

- defined on a interval $[a, b]$ with $a < b$
- the probability of any value in the interval is the same (outside is 0)

$$f(x) = U_{a,b}(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



- probability of an interval of length h : $\frac{h}{b-a}$

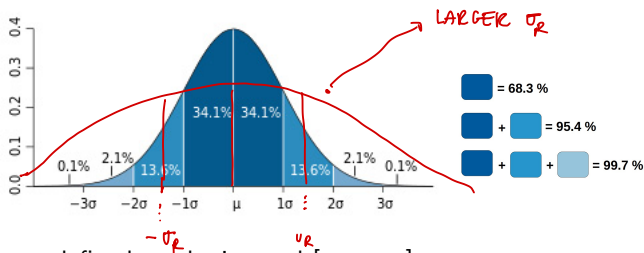
$$\int_y^{y+h} \frac{1}{b-a} dx = \frac{x}{b-a} \Big|_y^{y+h} = \frac{y+h-y}{b-a} = \frac{h}{b-a}$$

$$m = \mathbb{P}(x) = \frac{a+b}{2} \quad \int_a^b \frac{1}{b-a} dx = \frac{x^2}{2(b-a)} \Big|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)}$$

$$\text{VAR}(x) = \frac{(b-a)^2}{12} \rightarrow \boxed{\mathbb{E}(x^2) - \mathbb{E}(x)^2}$$

exercise 1.4

NORMAL DISTRIBUTION $N_{\mu,\sigma}(x)$



- defined on the interval $[-\infty, \infty]$

density:

$$f(x) = N_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

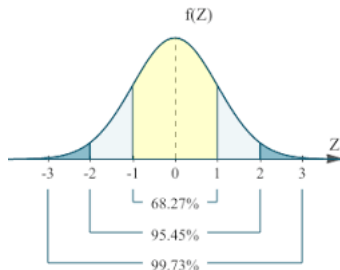
cumulative:

$$F(x) = \Phi_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} dy$$

- symmetric
- fully described by parameters μ (mean) and σ (standard deviation)
- central value = mean, median, mode



STANDARD NORMAL DISTRIBUTION $N(x)$ (often denoted as $N(Z)$)



- $\mu = 0$ and $\sigma = 1$

density:

$$N(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

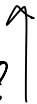
cumulative:

$$\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^Z e^{-\frac{1}{2}y^2} dy$$

- for $\Phi(Z)$ available a **table of values**

SYMMETRY $\rightarrow$ TABLES ONLY FOR POSITIVE VALUES

$P(Z \leq 1,95)?$



$$\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^Z e^{-\frac{1}{2}y^2} dy$$

0,7

0,5

Z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
+2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
+2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
+2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
+2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
+2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
+2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
+2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
+2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
+2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
+3	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900

$\Downarrow$
0,97441

$\cdot \bar{z}?$ /

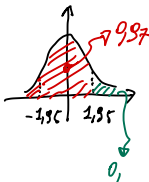
$P(Z \leq \bar{z}) = 0,9$

$\Downarrow$
1,285

$\cdot P(Z \leq -1,95)?$

$\Downarrow$

$1 - 0,97441$
 $\approx 0,0255$



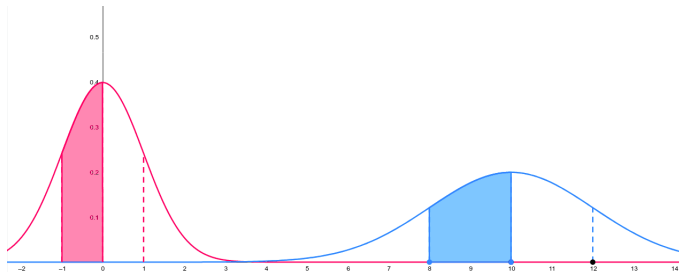
$\cdot \bar{z}$ /
 $P(Z \leq \bar{z}) = 0,3$

$1 - 0,3 = 0,7$

$\Downarrow$

-0,525

standardization of a normal variable



DISTRIBUTED LIKE...

↑
 $x \approx N_{\mu, \sigma}(x):$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

standardization

$$\Rightarrow Z = \frac{x - \mu}{\sigma} \Rightarrow$$

$$Z \approx N(0, 1):$$

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

$$\phi_{10, 2}(16) = \phi\left(\frac{16 - 10}{2}\right) = \phi(3)$$

it can be shown that

$$N_{\mu, \sigma}(x) = \frac{1}{\sigma} N\left(\frac{x - \mu}{\sigma}\right)$$

and

$$\Phi_{\mu, \sigma}(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$



CENTRAL LIMIT THEOREM (simplified)

given n random variables **independent** and **identically distributed** $X_1, X_2, \dots, X_n$,
their **arithmetic mean** tends to be normally distributed:

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n} \approx N_{\mu, \sigma}(x) \quad \text{as } n \rightarrow \infty$$

ALSO FOR SMALL n

also their **sum**:

$$S_n = X_1 + X_2 + \dots + X_n \approx N_{\mu, \sigma}(x) \quad \text{as } n \rightarrow \infty$$